

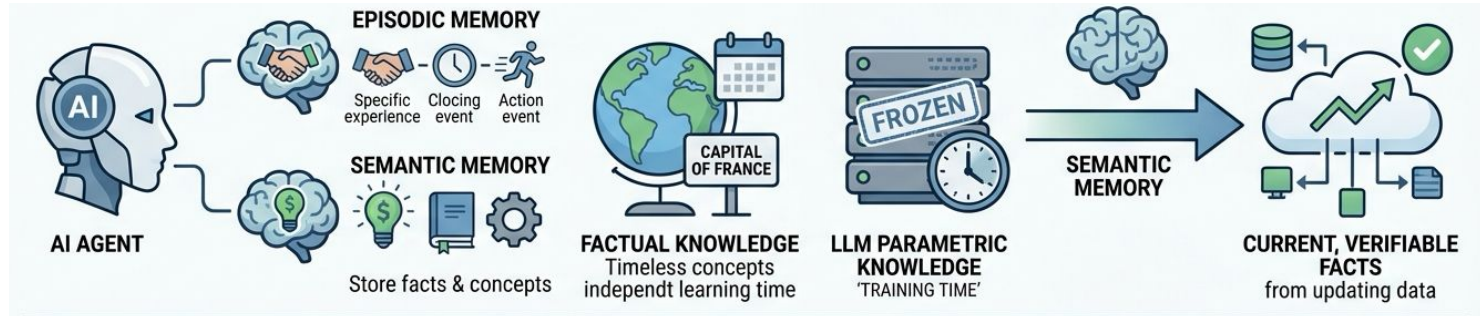


U.S. General Services
Administration

Chapter 5.8: Semantic Memory

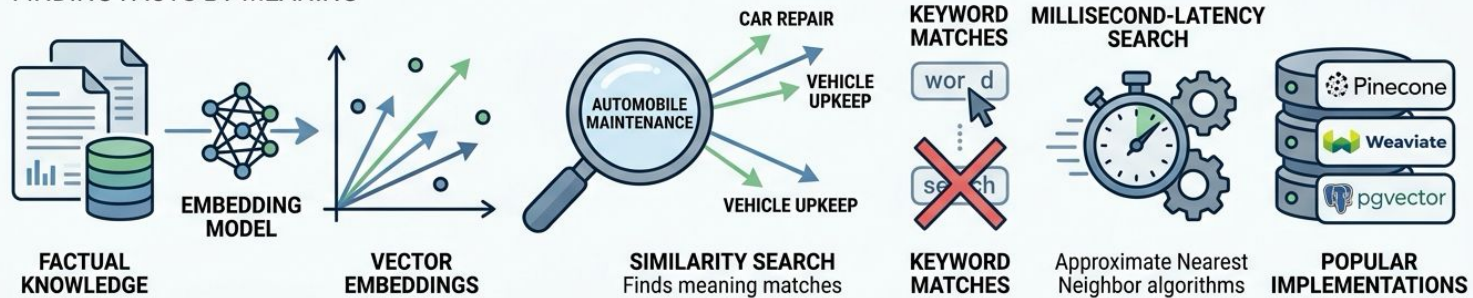
US General Service Administration
AI Community of Practice - March 2026

Vector Databases -- Finding Facts by Meaning

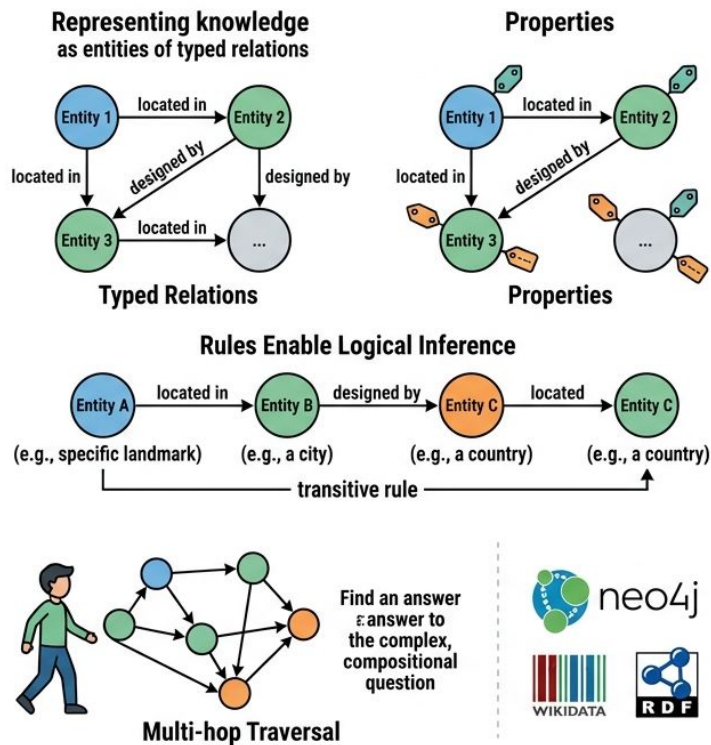


VECTOR DATABASES FOR MEANING-BASED RETRIEVAL

FINDING FACTS BY MEANING

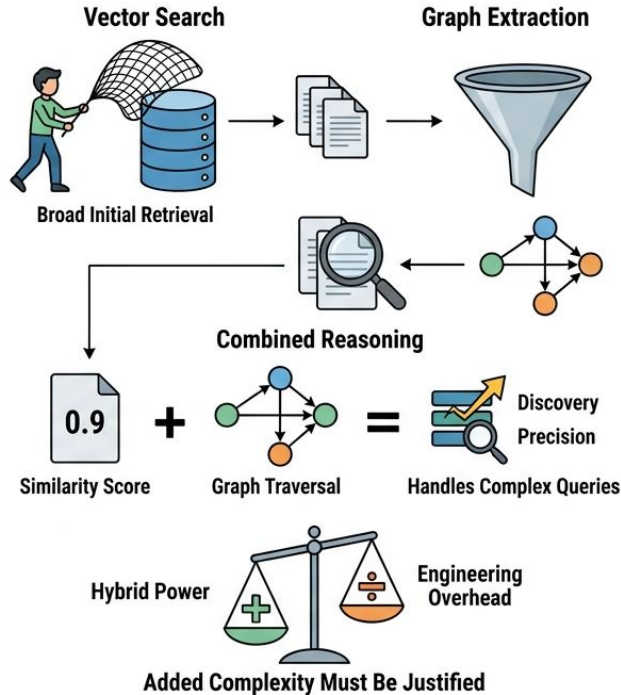


Knowledge Graphs -- Reasoning Through Relationships



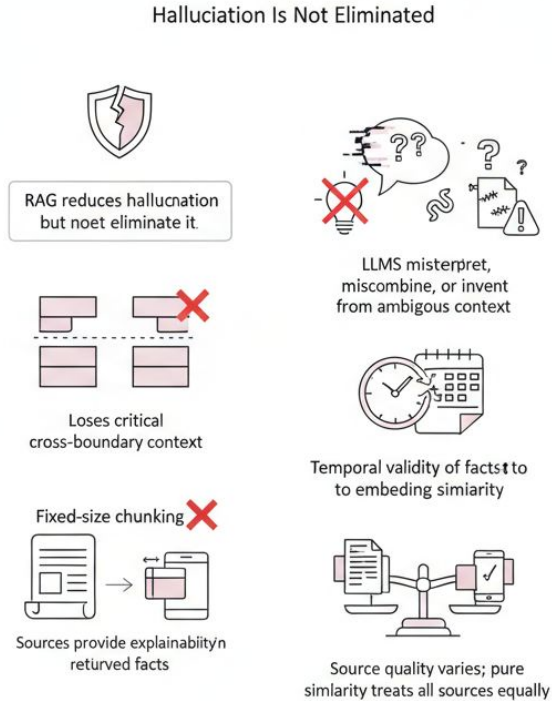
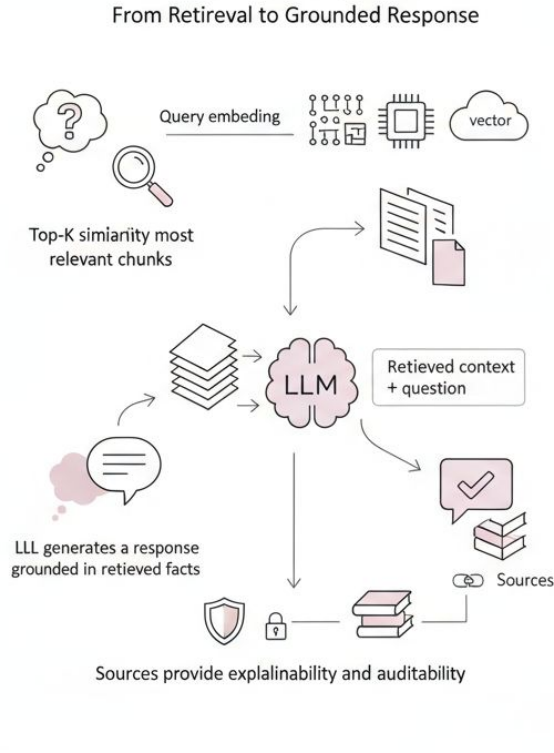
- Represent knowledge as entities connected by typed relations
- Properties attach attributes directly to entity nodes
- Rules enable logical inference beyond stored facts
- Multi-hop traversal answers complex compositional questions
- Neo4j, Wikidata, and RDF stores are common platforms

Hybrid Architectures -- Best of Both Worlds

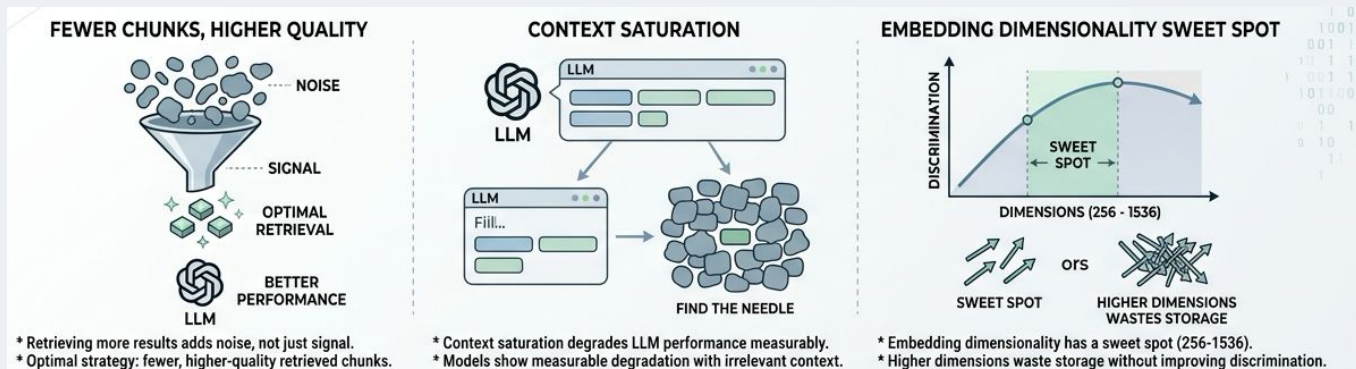


- Vector search performs broad initial retrieval across large bases
- Graph extraction identifies entities and relationships in results
- Combined reasoning merges similarity scores with graph traversal
- Handles complex queries needing both discovery and precision
- Added complexity must be justified by requirements

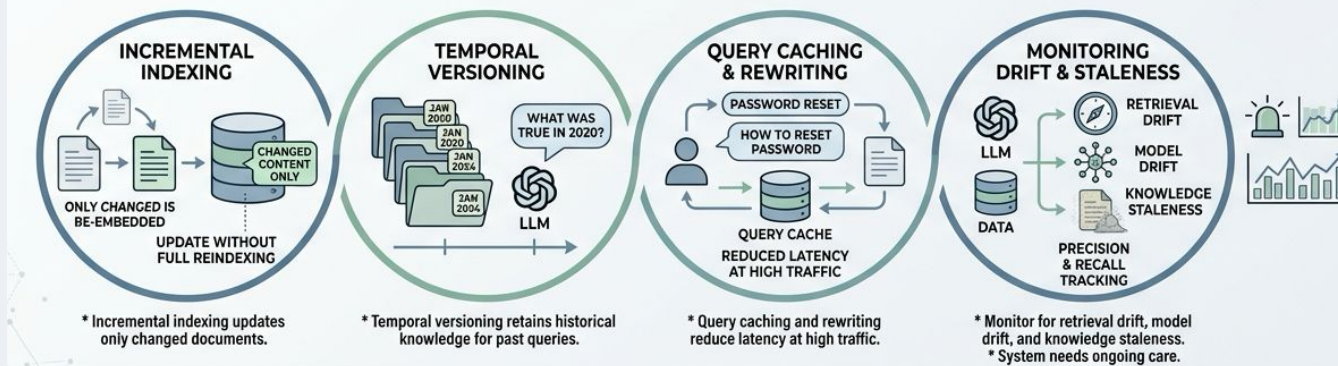
The RAG Pipeline & pitfalls



Retrieval Optimization -- Less Is Often More



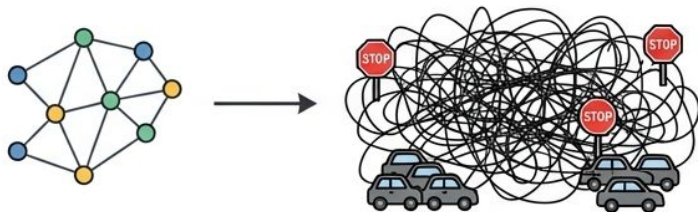
PRODUCTION SCALING -- KEEPING SEMANTIC MEMORY FRESH



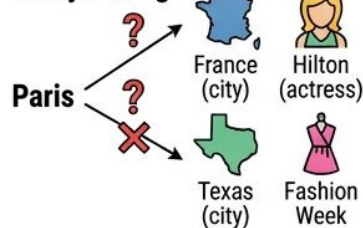
Knowledge Graph Challenges at Scale

Million-node demos

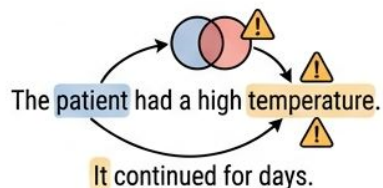
billion-node **production bottlenecks**



Entity linking



Coreference resolution



Knowledge uncertainty
probabilistic representation



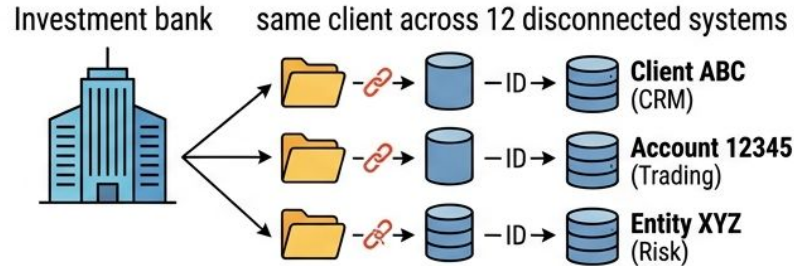
Drug (60-80% efficacy)
Drug (60-80% efficacy)
Drug (20% effc.)
Drug (10-50%)

Most graph databases lack
native probabilistic support

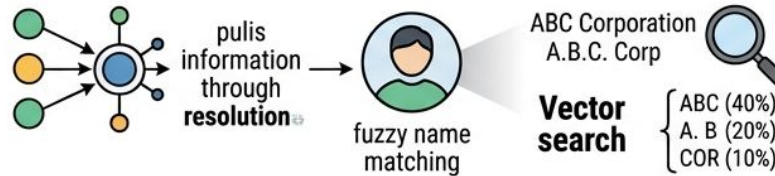


- Million-node demos hide billion-node production bottlenecks
- Entity linking fails on ambiguous terms like "Paris"
- Coreference resolution across sentences remains error-prone
- Knowledge uncertainty needs probabilistic representations
- Most graph databases lack native probabilistic support

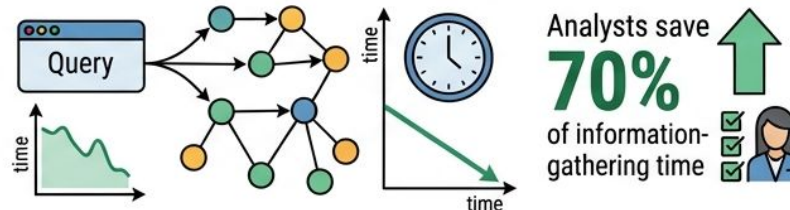
Case Study -- Unifying Fragmented Knowledge



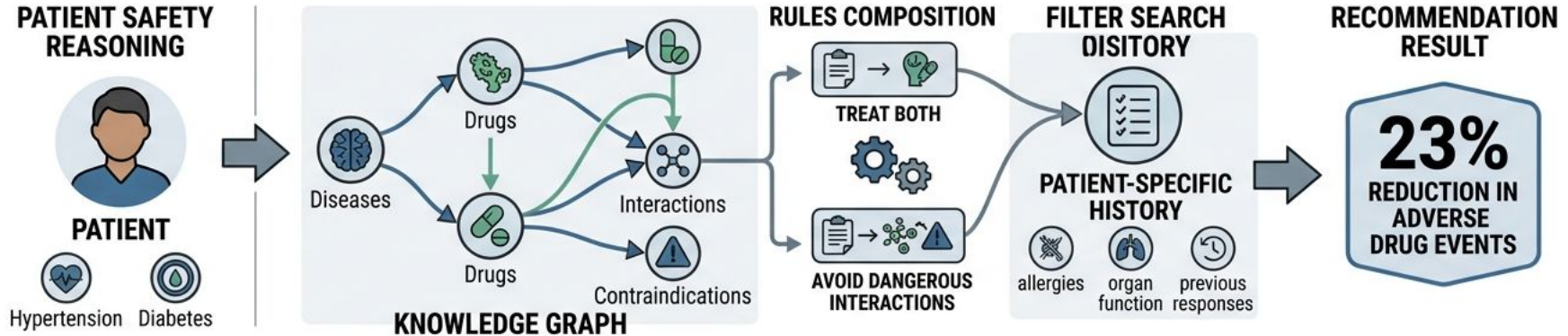
Knowledge graph unifies identities through **entity resolution**



Graph traversal answers complex cross-system queries instantly

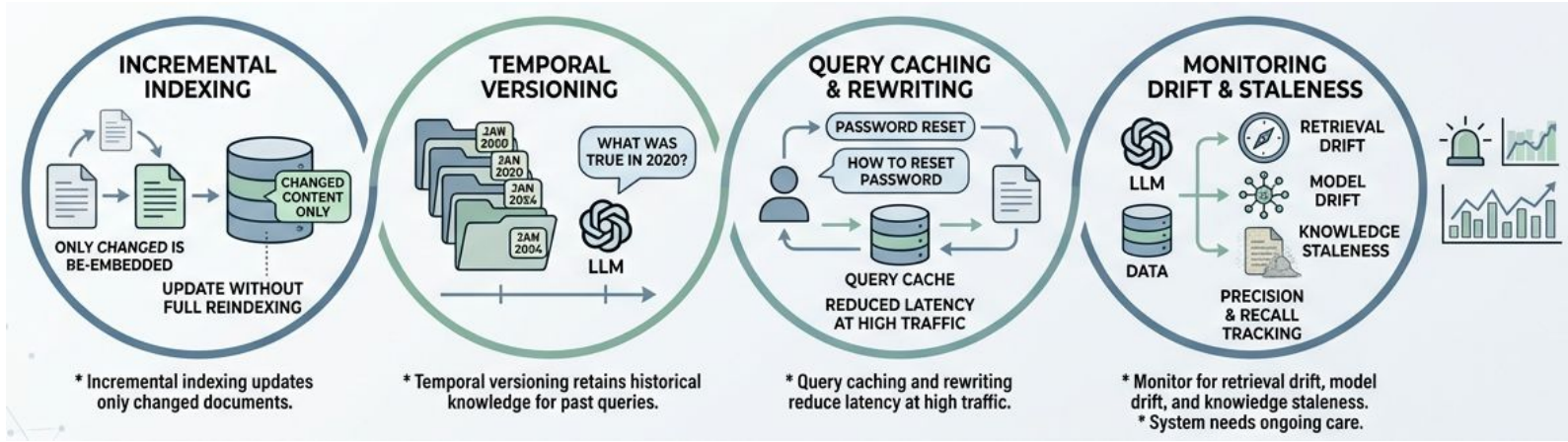


Case Study -- Multi-Hop Clinical Reasoning



- Patient with hypertension and diabetes needs safe medications
- Graph traverses diseases, drugs, interactions, and contraindications
- Rules compose constraints: treat both, avoid dangerous interactions
- Patient-specific history narrows theoretically safe options further
- Adverse drug events decreased 23% after deployment

Keeping Semantic Memory Fresh



- Full reindexing of millions of documents is cost-prohibitive
- Incremental indexing updates only changed documents
- Temporal versioning retains historical knowledge for past queries
- Query caching and rewriting reduce latency at high traffic
- Monitor for retrieval drift, model drift, and knowledge staleness

Key Takeaways

- Semantic memory stores factual knowledge independent of experience
 - Vectors excel at broad semantic discovery; graphs at logical precision
 - Hybrid architectures combine both when complexity justifies cost
 - Production demands incremental updates, caching, and drift monitoring
 - Next chapter: working memory manages what the agent holds right now
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